

andrew cotter

creative technology | design engineering

contact

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Professional Experience

2022 – 2025

Senior Creative Technologist @ The LAB At Rockwell Group

- Technical lead on four concurrent project scopes, owning frontend architecture and full-stack development from concept through fabrication and live install
- Built real-time React / Node.js / WebSocket systems for large-scale permanent installations serving hundreds to thousands of daily visitors
- Architected Three.js and WebGL rendering pipelines; optimized a Unity build to run at 4x the required framerate, enabling the full server stack to run on shared hardware and significantly reducing client hardware costs
- Developed VR previz environments in Unity, allowing clients to walk through spaces before fabrication began, reducing revision cycles on multi-million dollar projects
- Established 3D asset pipelines from CAD and Blender to web-ready and real-time formats, enabling designers to hand off directly to interactive platforms
- Drove R&D initiatives in mixed reality and sustainability technologies, informing studio proposals and expanding capabilities
- Mentored junior and intern technologists; led onboarding on team tech stacks
- Maintained warranties on three legacy projects— keeping installations running and answering clients' maintenance questions

2021 – 2022

Creative Coder @ Meta

- Shipped approximately a dozen AR templates, filters, and effects for Instagram and Facebook Stories, collectively reaching millions of users across Meta platforms
- Optimized assets and experiences for consistent performance across a wide range of mobile devices and screen sizes
- Prototyped new integrations between Stories and emerging Meta platform features including Avatars
- Operated in a high-velocity, high-deployment-frequency environment with a strict bar for cross-platform performance and visual quality

2017 – 2021

Freelance Creative Technologist

- Delivered bespoke interactive web and real-time media experiences for clients including Google, Hulu, Smithsonian Institution, Active Theory, Red Paper Heart, and Local Projects
- Designed and built custom networking, A/V, and real-time graphics systems for live and permanent installations
- Operated independently across diverse client contexts, consistently delivering under tight creative and technical constraints

2014 – 2016

Interaction Developer @ Night Kitchen

- Developed and maintained web and physical interactive experiences for museums and cultural institutions, including the Smithsonian Asian Pacific American Center, Chemical Heritage Foundation, Mutter Museum, and the University of Pennsylvania
- Researched and evaluated emerging web, media, and exhibit technologies; delivered prototypes and technical recommendations to inform team workflows

Teaching Experience

2018 - 2020

Adjunct @ Parsons Design & Technology

Classes: Creativity and Computation, Creative Coding with Unity, Immersive Storytelling

2021 – Present

Adjunct @ NYU Integrated Design & Media

Classes: Creative Coding, Advanced Creative Coding with WebGL

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Education

2018

Parsons School Of Design | The New School
MFA – Design & Technology

2014

The University Of The Arts
BS – Industrial Design

Skills

Front End Dev

- HTML, CSS, JavaScript, Typescript
- Frameworks (React, Vue, Next, Astro)
- WebGL (Three.js, React Three Fiber, Pixi.js, etc)
- Animation (GSAP, React Spring, Framer Motion, etc)
- Protocols/APIs (REST, GraphQL, Websocket)
- iOS Development (Swift, UIKit, SwiftUI, ARKit)

Systems & Devops

- JS Runtimes (Node.js, Electron, Deno, Bun)
- Version Control (Git, Github, LFS, Gitflow, etc)
- Protocols/APIs (REST, GraphQL, Websocket)
- Linux, Docker, TrueNAS Scale, AWS
- Rust (Nannou, Bevy, WASM, WebGPU)

Assets Pipelines & Design Tools

- Fluent in asset hand-off and optimization from:
 - 2D tools: Adobe CC, Figma, etc
 - 3D tools: Blender, Rhino, Maya, Spline

Immersive, Real-Time Media

- Unity3D (C#, VR/AR)
- TouchDesigner (Python)
- openFrameworks/Cinder (C++)
- Unreal Engine (Blueprints, C++)
- Shaders (GLSL, HLSL, Shader Graph, Metal, etc)

Physical Computing & A/V

- Arduino, Teensy, Raspberry Pi, and other PCB Micro-Controllers
- Circuit Prototyping, Soldering, Wiring
- Lighting/Audio Protocols (e.g. Serial, Midi, DMX, OSC)
- Audio DSP and Signal Processing

AI Tools

- LLMs (Claude, Claude Code, ChatGPT, Codex, Studio LM)
- Diffusion Models (via ComfyUI and Automatic1111)
- AI-First Editors/Orchestrators (Zed, Cursor)

Projects

2022-2025

Word Cloud, Milken Center For The Advancement Of The American Dream

Large-scale permanent installation in Washington DC. Visitors contribute words describing what the American Dream means to them; input feeds a live evolving 3D visualization in real time.

Full-stack architecture: Unity, Node.js WebSocket server, Next.js frontend, all running on shared hardware through framerate optimization.

MCAAD has been awarded *USA Today's* No. 6 Best New Museum of 2025.

2020

Huluween, The Screamlands

Interactive, WebGL-based haunted house for Hulu's annual Halloween campaign. Implemented dynamic LOD asset selection served from the back-end and meticulously packed texture atlases to maintain smooth performance across desktop and mobile at scale. Implemented 3D UI features, shaders for VFX, and a sound engine for narration and foley playback.